

Strategy Brief #3

Regional coastal land Exposure – Eastern Gulf of Guinea



Mapping how the combined forces of sea-level rise, land subsidence, and coastal erosion are placing large areas of economically valuable land at permanent risk of inundation, and why today's land-use decisions will define tomorrow's risk trajectories.

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<https://chounnagerginoy.github.io/>

Introduction

Coastal land in the region is increasingly exposed to environmental hazards due to **sea-level rise, erosion, and subsidence.**



Key Challenge

Large portions of economically valuable land are at risk of **permanent or recurrent inundation.**

Strategic Insights

- Subsidence significantly increases **effective sea-level rise**.
- Urban expansion is occurring in **highly vulnerable coastal zones**.
- Natural buffers (wetlands, mangroves) are **degrading rapidly**.

Strategic Implications

- Loss of land threatens **economic development and food systems.**
- Increased need for **costly protection or relocation strategies.**
- Land-use decisions today will determine **future risk trajectories.**

Recommendations

- Protect and restore **natural coastal buffers**.
- Implement **coastal zoning regulations**.
- Integrate **subsidence** into **land exposure assessments**.
- Develop **long-term coastal adaptation strategies**.

Value Proposition

Strategic land management reduces **future economic losses** and **supports sustainable development**.